

Lecture Material #9

Information Security

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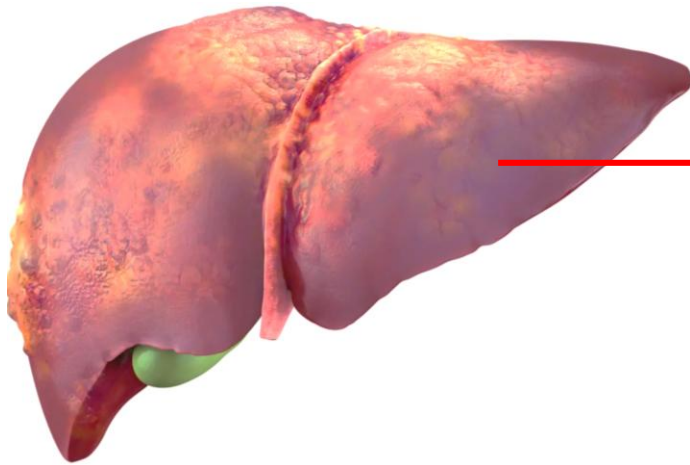
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Basic Data Statistics

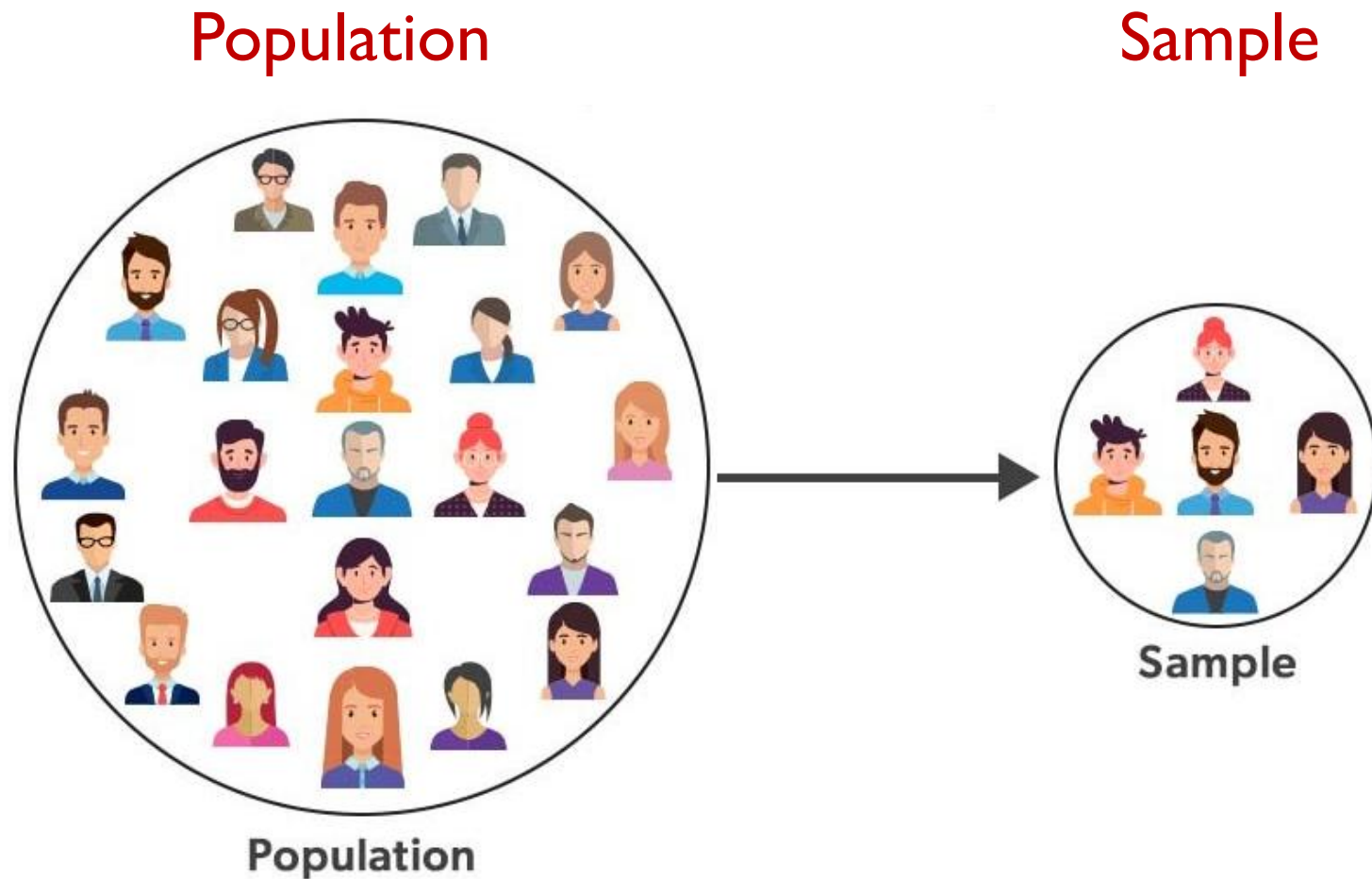
Population

Sample



240 Billion Cells

Basic Data Statistics



Basic Data Statistics

Population



Selection process



Sample



Research / Analysis done on
samples and not on populations

Basic Data Statistics

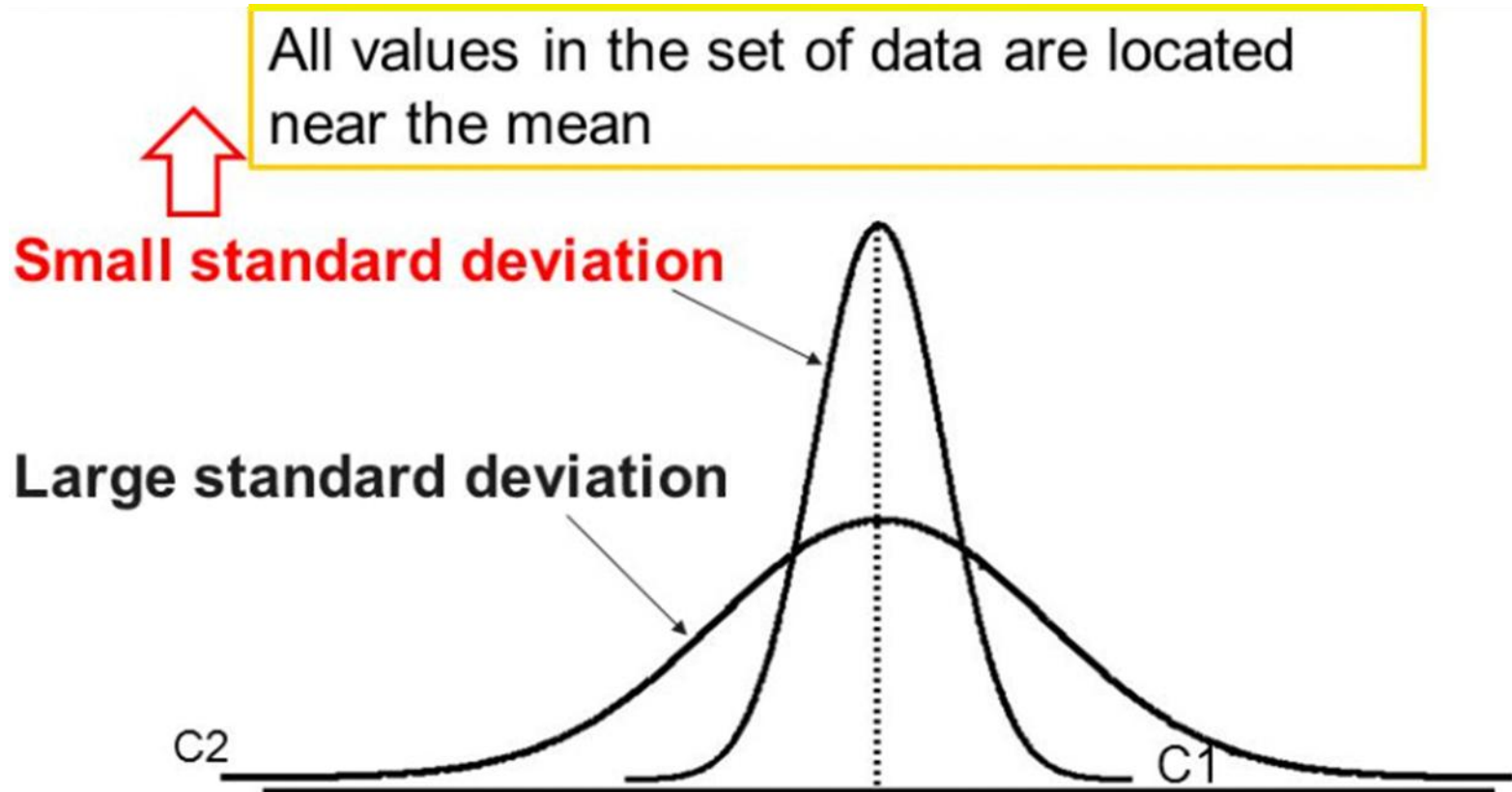
Standard Deviation:

The Standard Deviation is a measure of how spread out numbers are.

Variance:

The average of the squared differences from the Mean.

Basic Data Statistics



Basic Data Statistics

Population

$$\sigma = \sqrt{\frac{\sum (x_i - \mu)^2}{N}}$$

x_i = elements in population

μ = population mean

N = population size

Sample

$$s = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n - 1}}$$

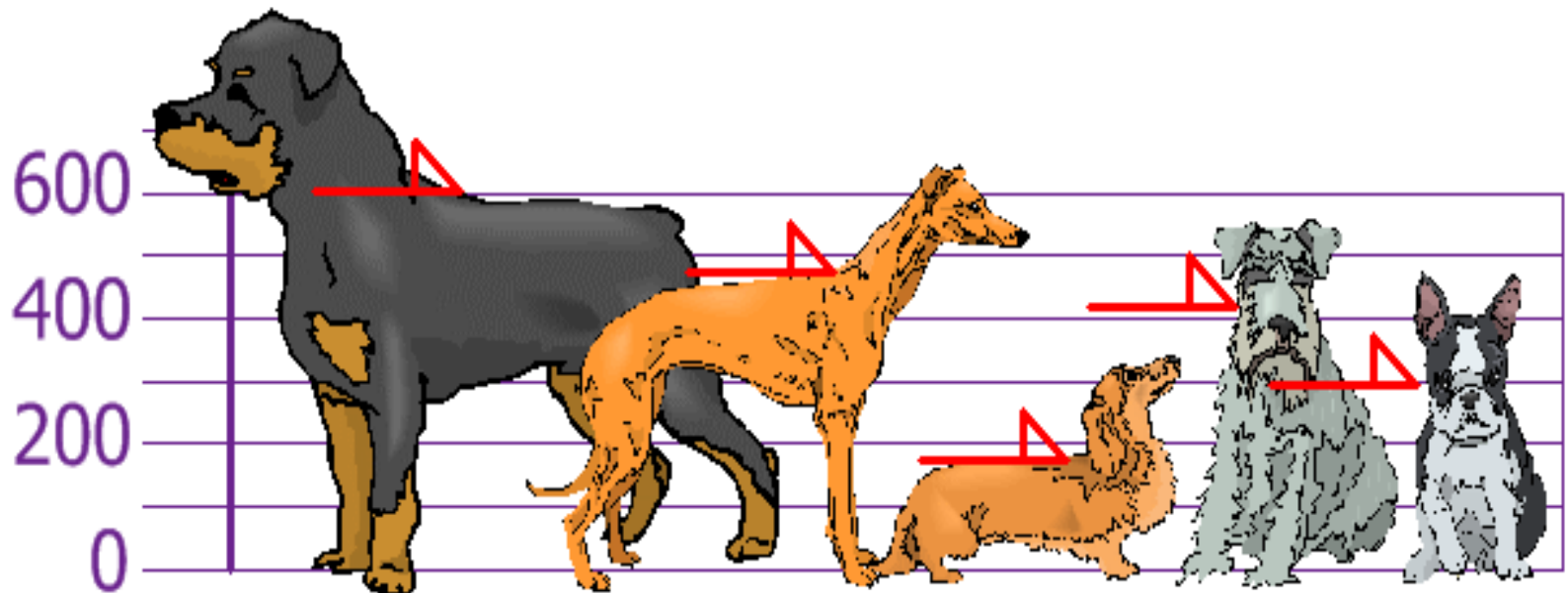
x_i = elements in sample

$\bar{x}$ = sample mean

n = sample size

Basic Data Statistics

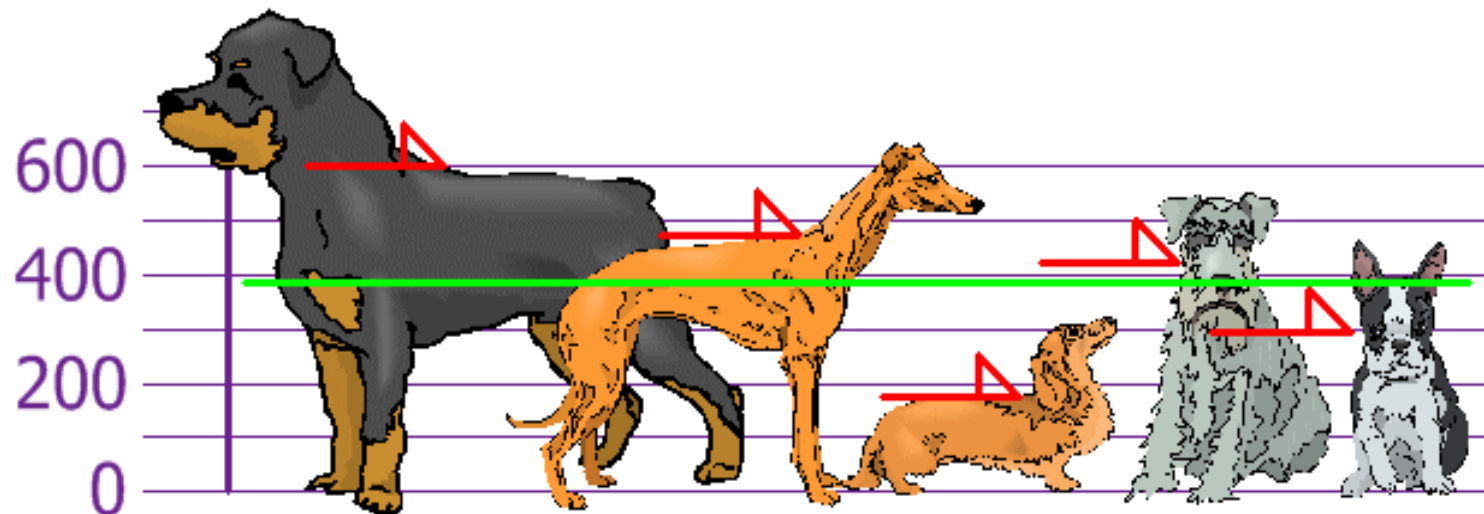
You and your friends have just measured the heights of your dogs (in millimeters): 600mm, 470mm, 170mm, 430mm and 300mm.



Basic Data Statistics

Mean:

$$\begin{aligned}\text{Mean} &= \frac{600 + 470 + 170 + 430 + 300}{5} \\ &= \frac{1970}{5} \\ &= 394\end{aligned}$$

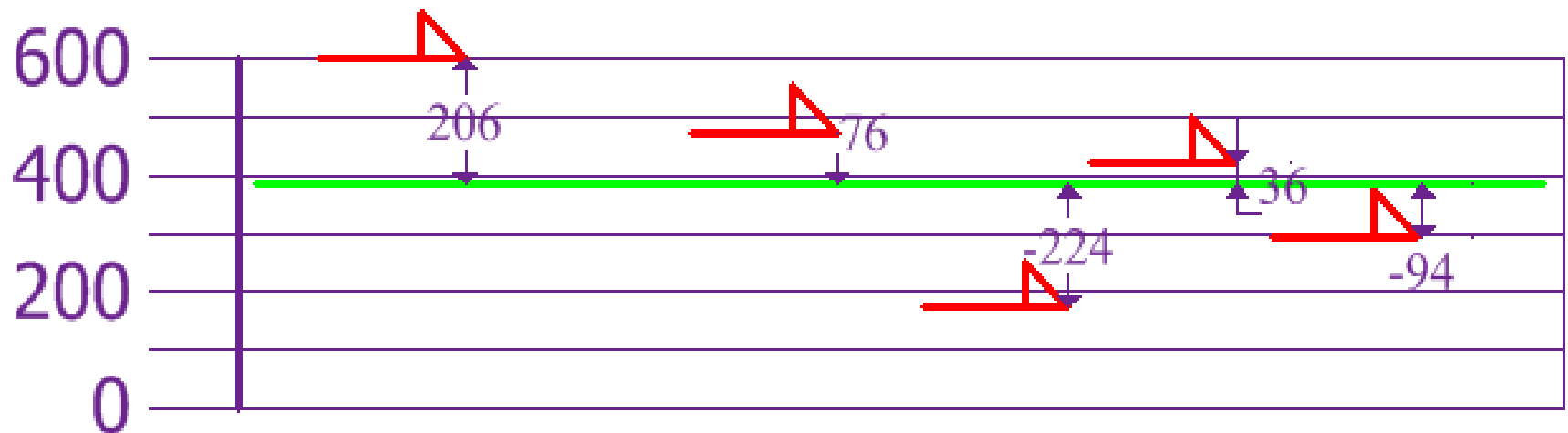


Basic Data Statistics

Variance:

To calculate the Variance, take each difference, square it, and then average the result:

600mm, 470mm, 170mm, 430mm and 300mm.



Basic Data Statistics

Variance:

To calculate the Variance, take each difference, square it, and then average the result:

Variance

$$\begin{aligned}\sigma^2 &= \frac{206^2 + 76^2 + (-224)^2 + 36^2 + (-94)^2}{5} \\ &= \frac{42436 + 5776 + 50176 + 1296 + 8836}{5} \\ &= \frac{108520}{5} \\ &= 21704\end{aligned}$$

Basic Data Statistics

Standard Deviation:

the Standard Deviation is just the square root of Variance:

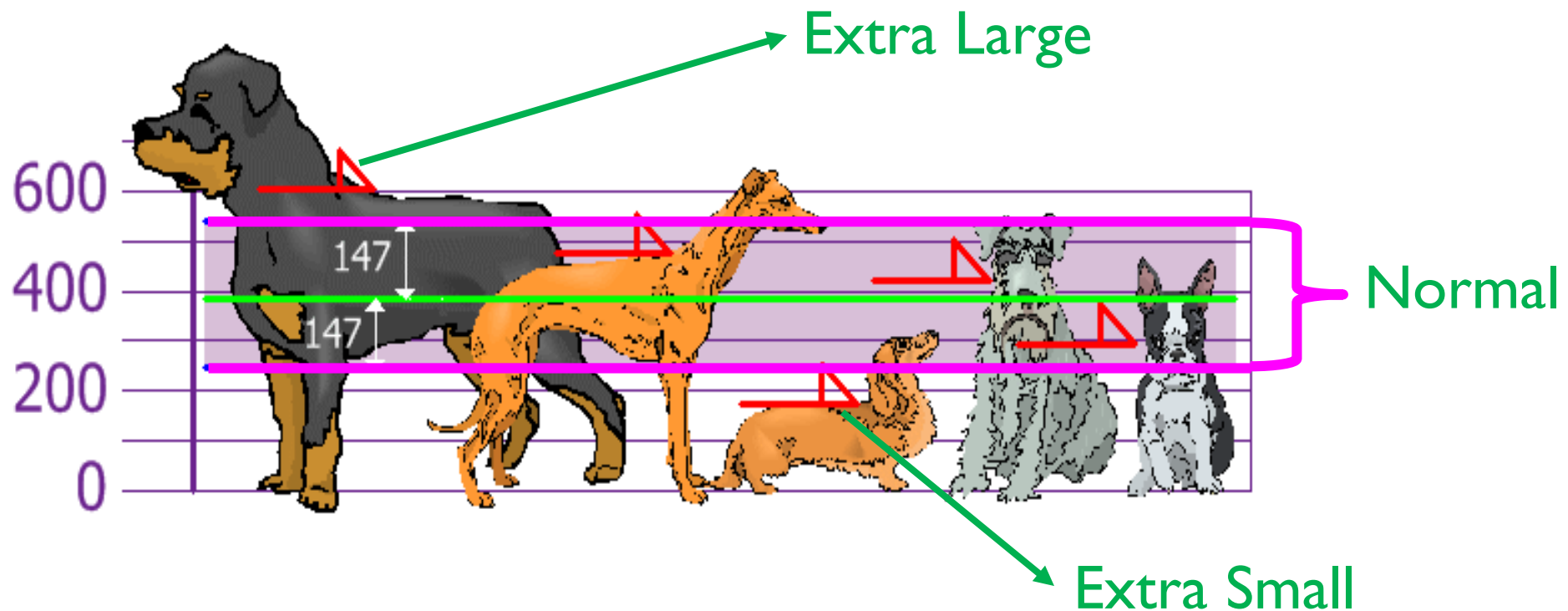
Standard Deviation

$$\begin{aligned}\sigma &= \sqrt{21704} \\ &= 147.32... \\ &= \mathbf{147} \text{ (to the nearest mm)}\end{aligned}$$

Basic Data Statistics

600mm, 470mm, 170mm, 430mm and 300mm.

Standard Deviation:



Basic Data Statistics

A news website categorizes its content into five sections: A, B, C, D, and E. The organization monitors the average weekly user engagement (in minutes) for each category as follows:

Category	A	B	C	D	E
Avg. Weekly Time Spent	146	38	102	7	89

In the context of information security risk management, user engagement is considered a proxy for asset value and potential business impact.

Classify each category into **critical, medium, and low-value information assets** based on the engagement data.

Basic Data Statistics

You are given a weekly sales rate of the following products. As a data analyst you need to decide the production rate of the products that needs to be increased or decreased or stopped.

Products	Weekly Sales Rate
A	110
B	15
C	20
D	70
E	3
F	15
G	40
H	30



